

EFFECTS OF RESISTED SPRINT TRAINING ON ACCELERATION IN PROFESSIONAL RUGBY UNION PLAYERS

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ABSTRACT

West, DJ, Cunningham, DJ, Bracken, RM, Bevan, HR, Crewther, BT, Cook, CJ, and Kilduff, LP. Effects of resisted sprint training on acceleration in professional rugby union players. *J Strength Cond Res* 27(4): 1014–1018, 2013—The use of weighted sled towing as a training tool to improve athlete acceleration has received considerable attention; however, its effectiveness for developing acceleration is equivocal. This study compared the effects of combined weighted sled towing and sprint training against traditional sprint training on 10 and 30 m speed in professional rugby union players ($n = 20$). After baseline testing of 10 and 30 m speed, participants were assigned to either the combined sled towing and sprint training (SLED) or traditional sprint training (TRAD) groups, matched for 10-m sprint times. Each group completed 2 training sessions per week for 6 weeks, with performance reassessed post-training. Both training programmes improved participants' 10 and 30 m speed ($p < 0.001$), but the performance changes (from pre to post) in 10 m (SLED -0.04 ± 0.01 vs. TRAD -0.02 ± 0.01 seconds; $p < 0.001$) and 30 m (SLED -0.10 ± 0.03 vs. TRAD -0.05 ± 0.03 seconds; $p = 0.003$) sprint times were significantly greater in the SLED training group. Similarly, the percent change within the SLED group for the 10 m (SLED -2.43 ± 0.67 vs. TRAD -1.06 ± 0.80 seconds; $p = 0.003$) and 30 m (SLED -2.46 ± 0.63 vs. TRAD -1.15 ± 0.72 seconds; $p = 0.003$) tests were greater than the TRAD group. In conclusion, sprint training alone or combined with weighted sled towing can improve 10 and 30 m sprint times; however, the latter training method promoted greater improvements in a group of professional rugby players.

KEY WORDS team sports, sprinting, speed

INTRODUCTION

Speed, or high maximum velocity, is frequently associated with successful performance in field sports (20). However, during competition, field sport athletes, such as soccer and rugby union players, rarely cover the necessary distance to achieve top speed. As such, the ability to accelerate, defined as the rate of change of velocity, is considered more fundamental to success in team sports rather than maximum velocity (4,6). In light of this, strength and conditioning researchers have focused on identifying suitable training methods to improve the acceleration abilities of such athletes.

Training methods to improve speed and acceleration have focused primarily on 2 main areas: assisted (3,7,15) and resisted sprint training (1,4,5,8,24). Assisted sprint training (e.g., downhill running) provides an effective stimulus for increasing stride frequency (15), whereas resisted sprint training (e.g., the use of a parachute, weighted vest, or sled) appears to promote greater neuromuscular activation and enhanced recruitment of the fast twitch muscle fibers (20), thereby increasing the propulsive forces generated by the leg musculature (1,14).

Recent research has produced conflicting results with regard to the effectiveness of resisted sprint training with Harrison and Bourke (8) reporting improvements with this training method, whereas Clark et al. (3) and Upton (20) reported no change in sprint performance with the application of resisted sprint training. Clark et al. (3) examined the effects of weighted sled or traditional speed training on time to approximately 18 and 55 m. Sled training had no effect on speed performance, whereas traditional sprint training did elicit some improvements. Harrison and Bourke (8) compared similar training methods and reported improved acceleration over 5 and 30 m with both approaches; however, the performance gains over 5 m was greater with sled training. These results suggest that traditional sprint training and resisted sled towing may both improve sprint times over distances more than 5 m, but at shorter distances (≤ 5 m), sled towing may provide a superior training stimulus.

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Many authors have suggested that resisted sled towing will not benefit sprint performance because of deteriorations in sprinting technique (13,17,19) and possible alterations in sprint mechanics (10,16,19). Taken together, it seems that weighted sled training may augment the force producing capability of muscle, but its transfer to performance is encumbered by a decline in sprinting kinematics if the load used is too great (14). Potentially, a combination of weighted sled and sprint training could provide the optimal stimulus for speed adaptations by developing muscle force and allowing sprinting technique to be maintained.

This study compared the effects of a combined weighted sled towing and speed training program against traditional speed training alone in a group of professional rugby union players. We hypothesized that both training programmes would lead to improvements in relevant measures of speed (10 and 30 m distances), but the performance changes with the combined training program would exceed that achieved by the speed training approach.

METHODS

Experimental Approach to the Problem

A group of professional male rugby union players were recruited to compare the effects of a combined sled towing and sprint training (SLED) program with a traditional sprint training (TRAD) program. The SLED involved 3 × 20 m sled tows with 12.6% body mass (14) and 2 series of 3 × 20 m sprints. The TRAD involved 3 × 20 m sprints and another 2 series of 3 × 20 m sprints. Both groups performed their respective training sessions twice a week for 6 weeks, and testing for 10 and 30 m sprint times was conducted pre- and post-training.

Subjects

With university ethical approval and informed consent, 20 professional rugby union players (Table 1) volunteered to take part in this study. Participants were recruited on the basis that they were healthy, injury free, and engaged in a structured weight training program for at least 2 years before the start of the study. The study started 2 weeks into the preseason period; before this, the players had been out of structured

TABLE 1. Participant details.*

Variables	SLED	TRAD	P
Weight (kg)	90.2 ± 10.3	90.9 ± 10.6	0.32
Stature (cm)	186 ± 8	185 ± 7	0.43
Age (y)	26.8 ± 3.0	25.1 ± 3.2	0.24
Predicted back squat 1 RM (kg)	186 ± 17	191 ± 15	0.52
10 m time (s)	1.74 ± 0.10	1.74 ± 0.07	0.97

*SLED = combined sled-towing and sprint training; TRAD = traditional sprint training; 1RM = 1 repetition maximum.

training for 4 weeks (however, all players were provided with a maintenance off-season program). Each participant was familiar with sled and traditional sprint training. Participants were assigned to either the SLED or TRAD programmes based on their 10-m sprint times, so that both groups were matched for 10 m speed (Table 1). Throughout the duration of the study, all participants engaged in 3 resistance training sessions (1 upper body, 1 lower body, and 1 overall), 3 conditioning sessions, 3 technical sessions, and 2 speed sessions (intervention).

Experimental Procedures

Ten and Thirty Meter Speed. Before the commencement of the training interventions, participants visited the laboratory for baseline testing of 10 and 30 m sprinting speed. Participants reported for baseline testing at 10 AM, having rested in the previous 24 hours, after consuming their typical training day breakfasts (replicated posttest) and having refrained from caffeine; moreover, participants had refrained from alcohol and strenuous exercise during the previous 24 hours. After the measurement of each participant's stature and body mass, participants underwent a standardized warm-up which comprised 10 minutes of movement-specific drills followed by 5 × 30 m sprints at submaximal pace (2 × 50%, 3 × 80% of maximum). After the warm-up, participants completed 3 × 30 m sprints from a stationary start (0.5 m behind

TABLE 2. Baseline and post-training 10 and 30 m times in the SLED and TRAD groups.*

	Baseline		Post-training	
	10 m (s)	30 m (s)	10 m (s)	30 m (s)
SLED	1.74 ± 0.10	4.26 ± 0.28	1.70 ± 0.10†	4.15 ± 0.28†
TRAD	1.74 ± 0.07	4.19 ± 0.19	1.72 ± 0.06†	4.15 ± 0.18†

*Data presented as mean ± SD. SLED = combined sled-towing and sprint training; TRAD = traditional sprint training.

†Indicates different to baseline ($p < 0.05$).

a predetermined start line), with full recovery between each trial (fasted 30 m time was recorded). Electronic timing gates (Brower TC-System; Brower Timing Systems, Draper, UT, USA) were placed at the start line and then 10 and 30 m distances from the first set of gates. All testing procedures were replicated post-training.

Training

Training was implemented during the preseason period; the intervention was placed such that it fell in a 6-week window where the players were not engaging in heavy contact sessions or preseason games. All training was performed on a flat surface, indoors, on a rubber crumb training field. Both training groups completed their respective programmes twice a week for a period of 6 weeks, with training taking place on Monday and Thursday mornings. Before each session, participants completed a standardized warm-up (same as pre- and post-testing), as prescribed by a certified strength and conditioning specialist. To ensure that the training session was equated, both groups completed a total of 240 m of towing or sprinting within each training session. Participants could consume water *ad libitum* during these training sessions.

Combined Sled Towing and Sprint Training. Participants completed 3×20 m sled tows, with 12.6% body mass (14) and 2 minutes recovery between efforts. According to Lockie et al. (14), a load of 12.5–13% of body mass is optimal as it does not alter sprint kinematics during the sled-towing exercise. After completion of the third 20-m tow, participants actively recovered for 8 minutes before completing 3×20 m sprints, with 2 minutes rest in between sprints. Once completed, participants repeated the above protocol.

Traditional Sprint Training. Participants completed 3×20 m sprints, with 2 minutes recovery in between sprints, and remained at rest for 8 minutes before completing another 3×20 m sprints, with 2 minutes rest in between sprints. Once completed, participants began an active recovery for 4 minutes before repeating the protocol. Both training programmes were implemented at the same time of day (10 AM), to account for diurnal variation in sporting performance, and the duration of the training sessions (approximately 36 minutes) were similar for both groups.

Statistical Analyses

Data are presented as mean $\pm$ SD. Within-group changes and between-group differences in sprint performance were analyzed using a 2-way (group $\times$ time) repeated-measures analysis of variance with Bonferroni adjustments and independent *t*-tests carried out where relevant. Statistical analyses were performed using SPSS software (version 16; SPSS, Inc., Chicago, IL, USA), with significance set at $p \leq 0.05$. Where statistical significance is indicated, 95% confidence

intervals (CIs) are presented for an estimate of the population mean difference.

RESULTS

Baseline 10- and 30-m sprint times were similar between the 2 groups at baseline (Table 2). There was a significant time effect ($p < 0.001$; partial $\eta^2 = 0.999$) but no effect of condition ($p = 0.696$) on sprint times after the 6-week training

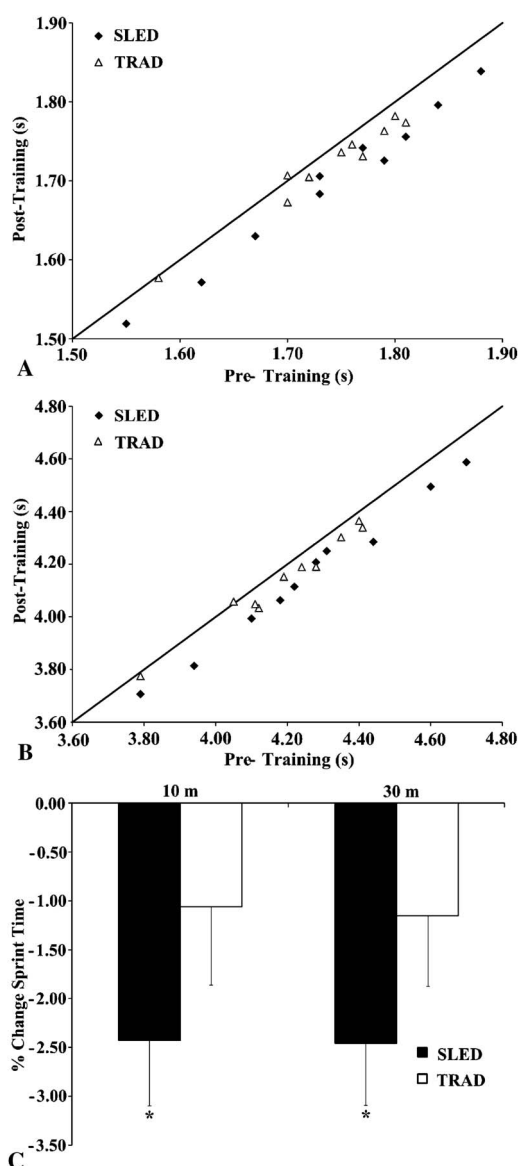


Figure 1. Individual changes in A) 10 and B) 30 m times, and the percentage change in 10 and 30 m speed, in the SLED and TRAD groups (C). Asterisk indicates significant difference by condition ($p < 0.05$). SLED = combined sled-towing and sprint training; TRAD = traditional sprint training.

period. After both training interventions, 10- and 30-m times significantly decreased from baseline (Table 2).

The individual responses to both training programs are presented in Figures 1A, B. When examining the changes in the participants' 10- and 30-m speed times, there were greater improvements in the SLED group than the TRAD group (Figure 1C). Changes (delta time) in both 10 m (SLED -0.04 ± 0.01 vs. TRAD -0.02 ± 0.01 seconds; $p < 0.001$; 95% CI = -0.003 to -0.027 seconds) and 30 m (SLED -0.10 ± 0.03 vs. TRAD -0.05 ± 0.03 seconds; $p = 0.003$; 95% CI = -0.014 to -0.084 seconds) times were greater for the SLED group. Similarly, when expressing the participants speed times as a percent change from baseline, the SLED group elicited the greatest improvements over the 10 m (SLED -2.43 ± 0.67 vs. TRAD $-1.06 \pm 0.80\%$; $p = 0.003$; 95% CI -0.91 to -2.23%) and 30 m (SLED -2.46 ± 0.63 vs. TRAD $-1.15 \pm 0.72\%$; $p = 0.003$; 95% CI -0.70 to -1.92%) distances, when compared with the TRAD group (Figure 1).

DISCUSSION

This study compared the effects of combined sled towing and sprint training against traditional sprint training alone on 10 and 30 m speed in a group of professional rugby union players. Our data demonstrated that employing 6 weeks of training (twice-weekly sessions) with either approach can improve acceleration over 10 and 30 m distances. However, the combination of sled tows with sprint training promoted greater gains in speed development.

We demonstrated that the SLED and TRAD approaches are both effective at improving speed over 10 and 30 m distances in professional rugby players. This finding is in agreement with Harrison and Bourke (8) who found that weighted sleds or traditional sprint training both promoted similar improvements in speed over 30 m. Conversely, Clark et al. (3) found weighted sled tows to be ineffective at improving speed over approximately 18 and 55 m, and they also demonstrated that traditional training provides a superior stimulus for speed development. This could be related to the sled training load employed. Clark et al. (3) used a sled weight of 10% of body mass, whereas Harrison and Bourke (8) used 13% of body weight, comparable to the 12.6% used in this study. According to Lockie et al. (14), the optimal sled load is 12.6% of body mass, as it will not unduly influence sprinting kinematics while still providing an effective overload stimulus. Potentially, the load used by Clark et al. (3) was simply too low. With regards to the speed improvements within this study, both training programmes may have induced increases in key determinants of acceleration and speed, such as peak horizontal and vertical impulses (9), peak force (21,22), and rate of force development (21).

Although both training groups experienced improvements in 10 and 30 m speed, the changes in sprinting performance from baseline was greater for the SLED group. The SLED program induced performance changes of -2.4% (10 m)

and -2.5% (30 m) compared to respective changes of -1.1% and -1.2% in the TRAD group. To our knowledge, we are the first study to demonstrate that combining weighted sleds with sprint training will improve speed more so than traditional sprint training alone. Also, the current findings were achieved within a group of professional athletes concurrently training for the sport of rugby union. We do acknowledge the possible contribution of other training factors or variables in concurrently training athletes, but all training factors (e.g., training mode, volume of load, exercise intensities) were consistent for both groups across this study.

The exact mechanism(s) underpinning the greater training effect in the SLED group is unclear. Speculatively, the 3×20 m tows, performed by the SLED group, may have acted as a preloading stimulus (2,11,12), inducing PAP during the 3×20 m sprints with chronic exposure leading to greater training responses. Although there is limited research to confirm or refute this hypothesis; at an acute level, research has demonstrated that inducing PAP, using a preload stimulus of heavy back squats (1 set of 3 repetitions at 91% 1 repetition maximum), can improve speed over 5 and 10 m (2). The observed differences may also be explained by other adaptations such as increased nerve conduction velocity and improved stretch reflex (18). Moreover, increases in muscular force output and, thus, the propulsive forces generated by the leg musculature, are important adaptations subsequent to sled training (1,14). In addition, it has been proposed that weighted sled training increases the eccentric strength of the leg extensor muscles during the braking phase of ground contact and an increase in muscle and leg spring stiffness, potentially decreasing ground contact time, and thus increasing stride rate (4,5,23). Furthermore, the SLED program may have induced greater improvements in peak horizontal force production, an important factor in acceleration (9). Collectively, the induction of these adaptations, combined with the possible maintenance of sprinting technique through the performance of traditional sprints, may have contributed to the greater improvement in speed seen within the SLED group.

Although it is important to take into account that these athletes were in preseason and thus there may be an element of a retraining effect in our data, this does not detract from the meaningfulness of the findings in that the speed was improved more so under the SLED condition. In conclusion, we have demonstrated that short distance speed can be improved through traditional sprint training alone, or combined with weighted sled tows, in a group of professional team sport athletes. However, greater improvements in speed were seen when employing a training program that combined weighted sled tows and sprint training.

PRACTICAL APPLICATIONS

The current findings have implications for athletes requiring accelerative abilities in sport. Specifically, sprint training appears to be effective in enhancing short distance (10–30 m)

sprinting speed in concurrently training athletes but more so when combined with weighted sled towing. The approach taken will be determined, among other factors, by the abilities of the athlete(s) to correctly perform sled-towing exercises and with additional loads. The results also suggest that twice-weekly training sessions performed during preseason, which focus on speed development, can benefit concurrently training athletes and in a relatively short period.

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